



VELS

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IMAGE PROCESSING ON SKIN DISEASES USING DEEP LEARNING

B.Tech CSE (Artificial Intelligent & Machine Learning)

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INTRODUCTION

DermaAI is a deep learning-based skin image classification system developed to analyze skin images and predict the corresponding skin lesion category. Skin diseases can have similar visual characteristics, making identification challenging without proper medical examination. The proposed system uses artificial intelligence and image processing techniques to simplify the analysis of uploaded skin images. It is developed using Python, TensorFlow/Keras, NumPy, PIL, and Streamlit. The application allows users to upload skin images in different formats, preprocesses the images by converting them to RGB format and resizing them to the required dimensions, and passes them to a trained deep learning model for class ification. The system identifies the category with the highest model output score and displays the predicted category, model score, and graphical representation of scores for all supported categories. The Streamlit interface provides a Simple and interactive environment for users to explore the classification results. The main objective of DermaAI is to demonstrate how deep learning can be applied to automated skin image classification and provide an accessible platform for educational and research purposes. However, the system is experimental and is not intended to replace professional medical examination or diagnosis.

ABSTRACT

Skin diseases are one of the most common health issues worldwide, affecting people of all age groups.
Early and accurate diagnosis is essential for effective treatment and to prevent further complications.
However, traditional diagnostic methods depend on expert dermatologists and can be time-consuming, costly and sometimes inaccessible, especially in remote areas. In recent years, deep learning techniques have shown great potential in medical image analysis. This project focuses on developing an image processing system using deep learning to detect and classify different skin diseases from dermoscopic and clinical skin images. The system uses convolutional neural networks (CNN) to learn important features from skin lesion images and classify them into various categories such as acne, eczema, psoriasis, fungal infection, and melanoma. The images are preprocessed using techniques like resizing, noise removal, contrast enhancement and data augmentation to improve the accuracy and robustness of the model. The trained model can accurately identify skin diseases and assist healthcare professionals in making faster and more reliable decisions. This approach provides a non-invasive, efficient and cost-effective solution for early skin disease detection, especially in areas with limited access to dermatologists.
It can also be integrated into mobile or web applications for real-time diagnosis and monitoring, Image processing combined with deep learning has the potential to improve dermatological care and create a healthier future for society.

PROBLEM STATEMENT

1. Difficulty in Skin Disease Classification: Skin lesions often have similar visual patterns, making it difficult to disting uish between different skin diseases using images alone.
2. Limitations of Manual Diagnosis: Manual skin image examination depends on dermatologists' expertise, can be tim e-consuming, and may be difficult to scale for large numbers of images.
3. Variations in Image Quality: Differences in image resolution, lighting, color, and image acquisition conditions can af fect the accuracy of skin disease classification.
4. Imbalanced Dataset: Public skin disease datasets contain unequal numbers of images across disease categories, which may cause the model to favor classes with more training samples.
5. Limited Feature Extraction: Traditional image-
processing techniques may fail to capture the complex visual patterns required to differentiate similar skin lesions.
6. Generalization Challenges: A model trained on a particular dataset may not perform consistently on images from di fferent skin tones, devices, or clinical environments.
7. Need for an Automated Classification System: There is a need for a reproducible system that preprocesses skin im ages, extracts meaningful features using ResNet-50, and classifies them using a Support Vector Machine (SVM).

PROJECT OBJECTIVES

- Collect and organize labelled skin-lesion images from a reliable public dataset.
- Preprocess images using resizing, normalization and quality enhancement.
- Use transfer learning / deep feature extraction with ResNet-50.
- Train a n SVM classifier o n the extracted deep features.
- Evaluate the model using accuracy, precision, recall, F1-score and confusion matrix.
- Build a simple prediction workflow for research and educational demonstration.

Literature Survey

Year and Paper/study	Method used	Key Findings	Limitations
2024 – SkinFormer (Xu et al.)	Transformer-based skin lesion segmentation with statistical texture representation.	Reported improved lesion segmentation results across public datasets, including ISIC 2018.	More complex architecture; segmentation performance does not directly establish disease-classification accuracy.
2025 – Hierarchical Skin Lesion Classification with Prototypcal Decision Tree npj Digital Medicine	Hierarchical prototypical learning combined with decision trees.	Supports interpretable predictions and considers relationships between disease categories; evaluated on 235,268 images covering 65 conditions.	Hierarchical classification is more complex than a conventional classifier and depends on the quality of the Disease hierarchy.
2026 – DermaScanAI	Multi-scale CNN features, Transformer encoders, attention mechanisms, and Grade-CAM explainability.	Reported 94.8% accuracy and 91.9% macro F1-score on HAM10000.	Evaluation on a benchmark dataset does not by itself establish reliability in clinical settings or across diverse populations.

Paper/study	Method used	Key Findings	Limitations
2026 – Hybrid Deep Feature Fusion and Ensemble Learning	ResNet-50 and DenseNet-201 feature fusion, PCA, SMOTE, and SVM/ensemble classification.	Reported 90.49% accuracy and an F1-score of 0.8489 on HAM10000. Also evaluated on ISIC 2019.	Multiple feature-processing and classification stages increase implementation complexity; independent clinical validation remains important.
2026 – CANet: Color-Aware	CNN with trainable, clinically in formed color representations.	Focuses on learning meaningful color patterns associated with skin lesions rather than treating color channels independently.	Color features alone may not capture all lesion characteristics; generalization across Image-acquisition conditions needs evaluation.

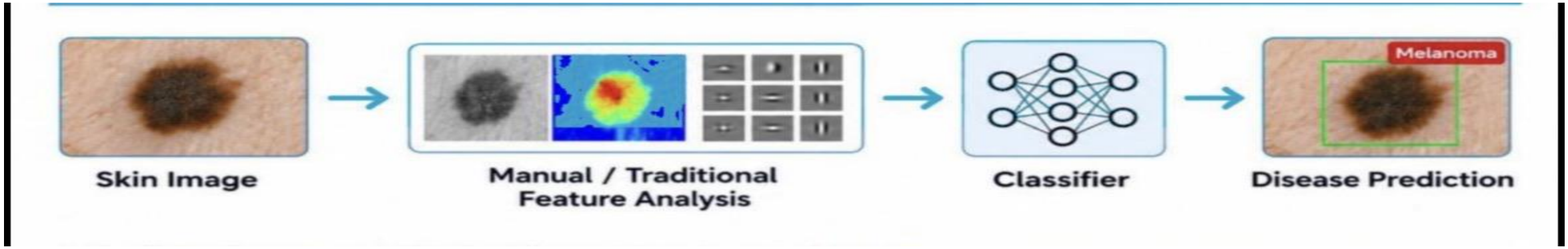
RESEARCH GAP

1. Class imbalance: Skin image datasets contain unequal numbers of images across disease categories, which can cause models to perform better on common diseases than on rare diseases.
2. Feature extraction and classification: Many approaches use a single deep learning model for classification. There is scope to investigate whether combining

ResNet-50 feature extraction with a Support Vector Machine (SVM) classifier improves classification performance.

3. Limited interpretability: A predicted disease label alone may not help users understand the model's output. Presenting prediction scores and category-
   level results can make the system's output easier to interpret, although these scores do not necessarily represent calibrated probabilities.
4. Generalization to unseen images: A model trained on a specific dermoscopic dataset may not perform equally well on images from different devices, skin tones, lighting conditions or clinical settings.
5. Practical accessibility: Some research models require technical knowledge or specialized software. A simple web in terface can make image-based classification easier to access and test.
6. Clinical limitations: Image classification alone cannot replace a dermatologist's examination, clinical history or biopsy when required.

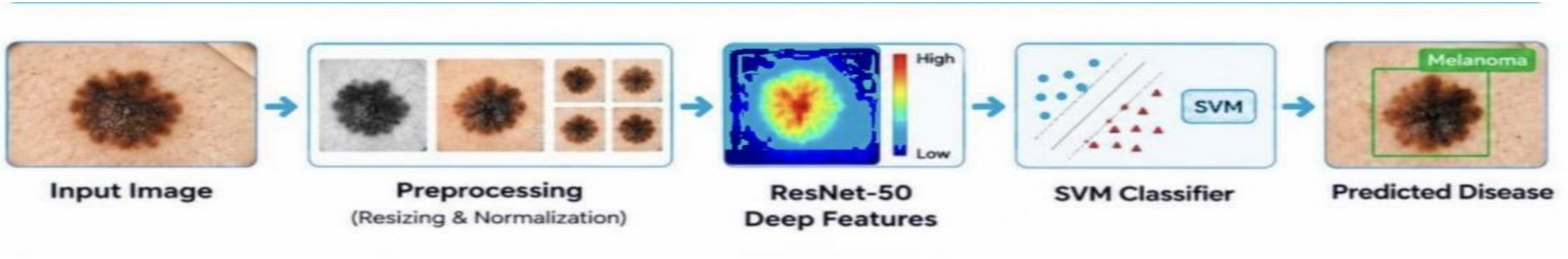
EXISTING SYSTEM



Limitaions of Existing Approaches

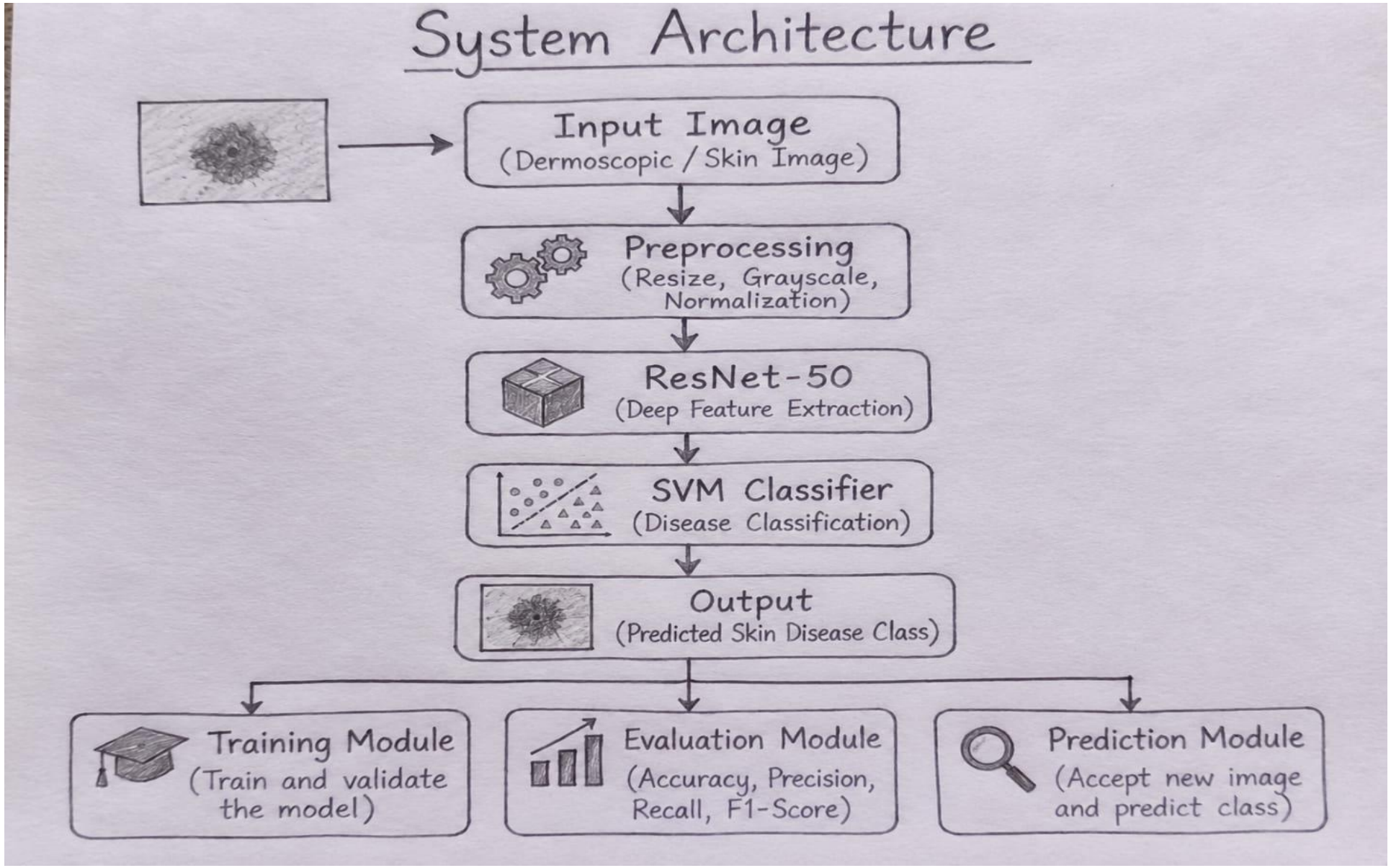
- Manual assessment is time-consuming and depends on expert availability.
- Traditional hand-crafted features may not capture complex lesion patterns.
- Dataset imbalance and acquisition differences can reduce generalization.
- Some published studies use private o r non-reproducible datasets.

PROPOSED SYSTEM



- Standardize input images using resizing and normalization to ensure uniformity.
- Apply image enhancement and augmentation techniques to improve model robustness and handle variations in the dataset.
- Use pre-trained ResNet-50 to extract discriminative deep features from the images.
- Train an SVM classifier using the extracted feature vectors for disease prediction.
- Report prediction confidence/score along with the class label for better interpretability.

SYSTEM ARCHITECTURE



METHODOLOGY

- Image Upload: Upload a skin image through the Streamlit web application.
- Image Preprocessing: Convert the image to RGB format and resize it to 224 × 224 pixels.
- Input Preparation: Convert the image into a NumPy array for model input.
- Model Loading: Load the trained TensorFlow/Keras model.
- Image Classification: Predict the skin lesion category using the model.
- Result Generation: Identify the category with the highest model score.
- Result Visualization: Display the predicted category, model score, and chart of category scores.
- User Interface: Provide an interactive Streamlit application for image-based classification.

TECHNOLOGY USED

- Python: Core programming language.
- TensorFlow/Keras: Loading and running the trained deep learning model.
- Streamlit: Developing the interactive web application.
- NumPy: Image array processing and prediction score calculations.
- PIL (Pillow): Image loading, RGB conversion, and resizing.
- JSON: Storing and retrieving skin disease class names.
- Deep Learning: Classifying skin images into supported categories.
- HTML/CSS: Customizing the application's appearance through Streamlit styling.

DATASETS

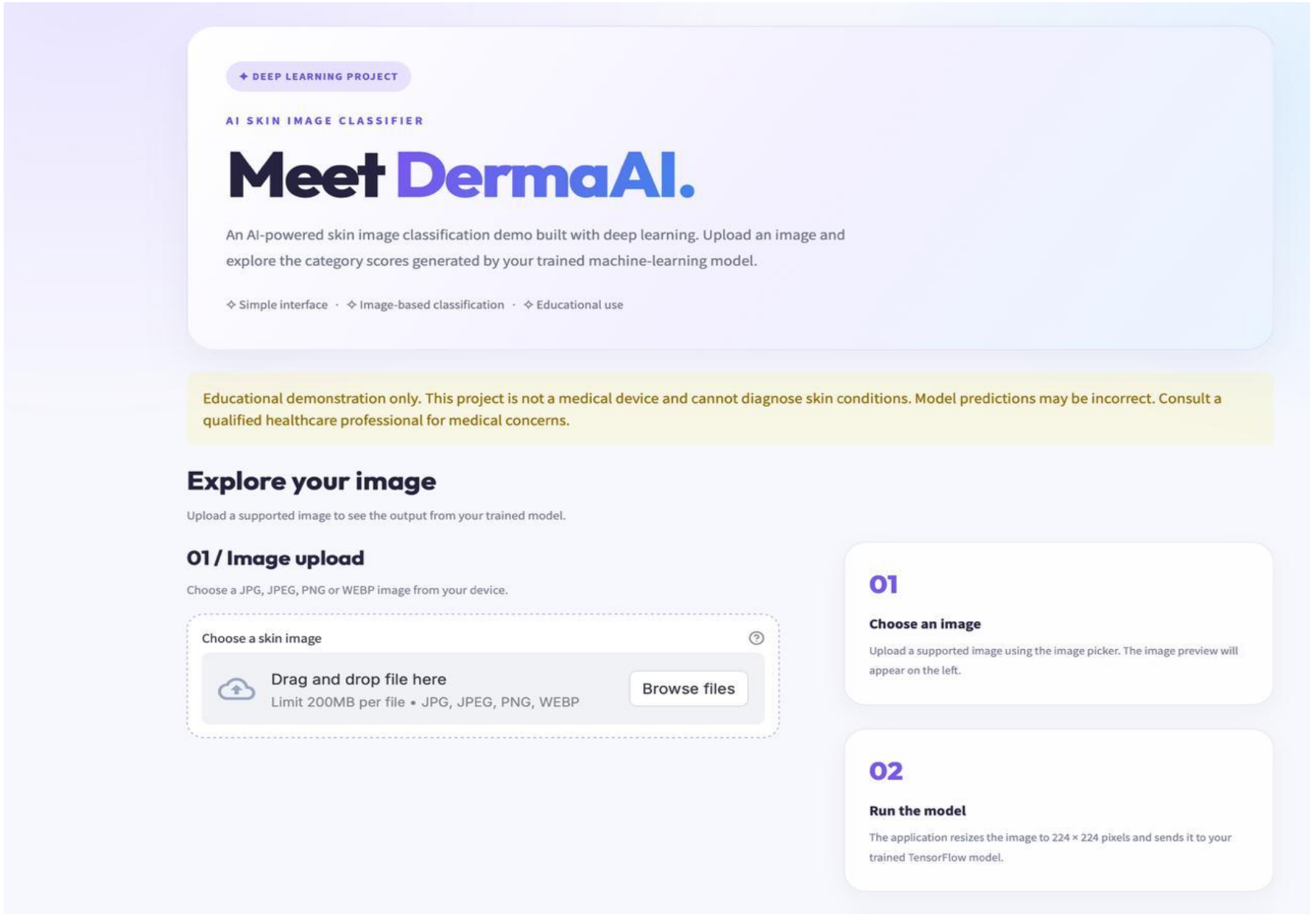
Dataset Used – DermaAI
HAM10000 Dataset (Human Against Machine with 10,000 Training Images)
The HAM10000 dataset is a publicly available collection of dermatoscopic images used for skin le sion classification and research in dermatology.

- Dataset Name: HAM10000 – Human Against Machine with 10,000 Training Images
- Total Images: 10,015 dermatoscopic images
- Image Type: Dermoscopic skin lesion images
- Number of Classes: 7
- Data Format: Images with corresponding diagnostic labels
- Application: Training and evaluation of skin lesion classification models

IMPLEMENTATION - MODULES

- Image Upload Module: Allows users to upload skin images in JPG, JPEG, PNG, or WEBP format.
- Image Preprocessing Module: Converts images to RGB and resizes them to 224 × 224 pixels.
- Deep Learning Model Module: Loads the trained TensorFlow/Keras model to analyze uploaded im ages.
- Classification Module: Predicts the skin lesion category based on the model's output scores.
- Result Analysis Module: Identifies the highest-scoring category and displays scores for all categories.
- Visualization Module: Presents classification results using metrics, charts, and detailed scores.
- Web Application Module: Integrates all components into an interactive Streamlit interface.

RESULT-CURRENT PROGRESS



01 / Image upload

Choose a JPG, JPEG, PNG or WEBP image from your device.

Choose a skin image

Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG, WEBP

Browse files

PAD_PAT_535_1011_567.jpg 24.7KB



Image selected for the model

✦ Run Classification

01

Choose an image

Upload a supported image using the image picker. The image preview will appear on the left.

02

Run the model

The application resizes the image to 224 × 224 pixels and sends it to your trained TensorFlow model.

03

Review the scores

View the highest-scoring model category and a chart of the scores returned by the classifier.

Your results

These are model-generated scores, not a confirmed medical diagnosis.

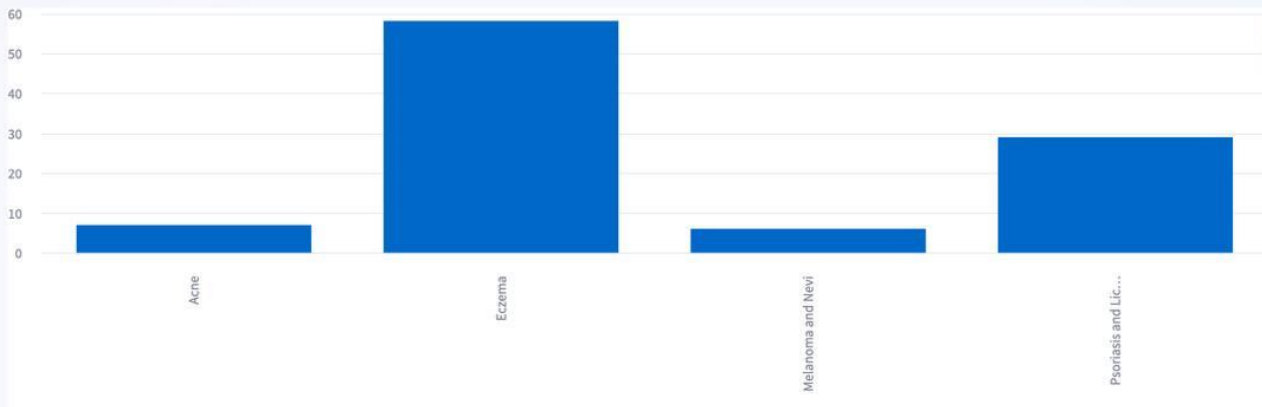
✦ TOP-SCORING MODEL CATEGORY

Eczema

Model score: **58.13%**

This score reflects the model output and must not be interpreted as diagnostic certainty.

Category score comparison



View detailed scores

Explore supported categories

These are the four labels configured in this project. They are model categories, not diagnoses.

FUTURE ENHANCEMENT

- Increase dataset diversity across skin tones, devices, age groups and clinical settings.
- Address class imbalance using balanced sampling, augmentation or suitable loss functions.
- Compare ResNet-50 + SVM with newer transfer learning architectures.
- Add explainability methods such as Grad-CAM to visualize image regions influencing predictions.
- Perform external validation o n a separate public dataset before claiming generalization.
- Develop a secure user interface and maintain privacy when handling medical images.
- If intended for clinical use, conduct proper clinical validation and regulatory review.

CONCLUSTION

Conclusion

• The project proposes an end-to-end image-processing and deep-learning pipeline for skin-lesion classification.
• By combining preprocessing, ResNet-50 deep features and SVM classification, the system provides a structured and measurable research prototype.

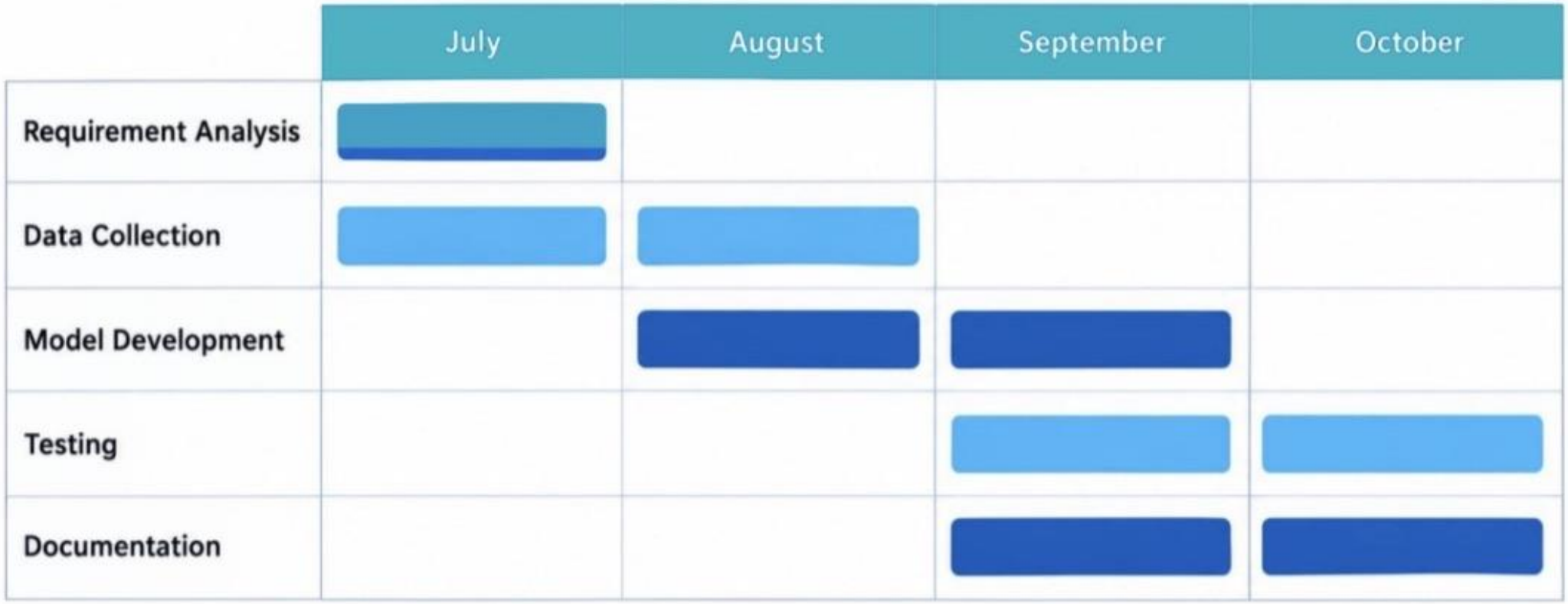
Key outcome

A reproducible workflow from skin image input to predicted disease category with quantitative evaluation.

Scope

Educational/research decision support only; not a substitute for dermatologist diagnosis

GANTT CHART - PROJECT PROGRESS (JULY-OCTOBER 2026)



Process plan: Requirement analysis->Data collection->Model development->Testing->Documentation(Update every week/ as per progress)

THANK YOU

Image Processing on skin Diseases Using Deep Learning

Mini Project – Final review